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Diagnostic accuracy of urinary *PENK* methylation test for urothelial and other cancers: A prospective study

Sangmin Lee, Bumjin Lim, Jungyo Suh, Bumsik Hong & In Gab Jeong✉

The methylated *PENK* gene is known to be highly expressed in the urine of patients with bladder cancer. Recent large-scale multicenter studies have confirmed the robust diagnostic performance of urinary *PENK* methylation, reporting high sensitivity and specificity for bladder cancer detection. However, its association with other cancers remains unclear. This prospective study aimed to determine the diagnostic accuracy of the urinary *PENK* methylation test for urothelial carcinoma among patients with 13 different cancer types. Between August 2022 and April 2023, 183 patients were enrolled, including 17 with urothelial carcinoma (eight bladder cancer and nine upper tract urothelial carcinoma) and 166 with other cancers. The urinary *PENK* methylation test showed an overall sensitivity of 94.1% (87.5% for bladder cancer and 100% for UTUC) and a specificity of 95.8% when patients with non-urothelial cancers were used as the control group. Positive urinary *PENK* methylation test was obtained for two patients with cervical cancer (25.0%), 2 with colorectal cancer (10.5%), and one patient each with esophageal (4.8%), liver (5.0%), and kidney (5.3%) cancer. These findings confirm the high sensitivity and specificity of the urinary *PENK* methylation test for detecting urothelial carcinoma and support its potential as a non-invasive biomarker.

Keyword DNA methylation, Hematuria, Tumor biomarkers, Urinary bladder neoplasms

Abbreviations

UTUC	Upper tract urothelial carcinoma
FDA	United states food and drug administration
IRB	Institutional review board
APCT	Abdominopelvic computed tomography
CI	Confidence interval
qMSP	Quantitative methylation-specific PCR
PCR	Polymerase chain reaction
NMP22	Nuclear matrix protein 22

Urothelial carcinoma, which encompasses bladder cancer and upper tract urothelial carcinoma (UTUC), represents a significant health burden due to its high prevalence and mortality rates. Bladder cancer is the 10th most diagnosed cancer globally, with approximately 573,000 new cases and 213,000 deaths recorded in 2020[1]. UTUC, which occurs in the renal pelvis and ureter, accounts for approximately 5% of all urothelial carcinoma cases[2]. Despite its lower incidence, UTUC generally has a more aggressive clinical course than bladder cancer, with approximately two-thirds of patients having invasive disease at diagnosis[3].

Accurate and early diagnosis of urothelial carcinoma is crucial yet challenging and often requires invasive procedures, such as cystoscopy and ureteroscopy. Cystoscopy, the current gold standard for the evaluation of suspected bladder cancer, is associated with significant limitations, including invasiveness, pain, and high cost[4,5]. Although ureteroscopy is valuable for evaluating UTUC, it is technically challenging and often yields limited tissue. It is also associated with potential complications, such as bleeding, tumor seeding, and ureteral perforation, and requires general anesthesia, which further complicates the diagnostic and treatment process[6]. These limitations reinforce the clinical need for reliable non-invasive diagnostic tools.

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Urine cytology, a widely used non-invasive detection method for bladder cancer, has a high specificity (86%) but limited sensitivity, particularly for low-grade tumors (as low as 16%)[7]. Several urinary assays approved by the United States Food and Drug Administration (FDA) are used alongside cystoscopy for diagnosis and surveillance. However, these biomarkers are limited by their low sensitivity[8,9].

In recent decades, DNA methylation has emerged as a promising biomarker for the detection of various cancers[10,11]. Among these, proenkephalin (*PENK*) methylation in urine has shown strong potential as a non-invasive biomarker for bladder cancer, with previous studies demonstrating its high diagnostic performance[12–14]. The *PENK* gene encodes a precursor protein that is enzymatically cleaved into enkephalins, active opioid peptides involved in cellular signaling. Hypermethylation of the *PENK* promoter may lead to gene silencing and has been proposed to inactivate tumor-suppressive pathways in certain cancers[15]. Recently, a multicenter study reported that a urinary DNA methylation test significantly improved the detection of bladder cancer, showing high sensitivity and specificity across different disease stages[16]. However, despite the well-documented use of urinary *PENK* methylation as a diagnostic marker for bladder cancer, its expression in patients with other cancer types, including UTUC, is less clear. Several studies have reported the hypermethylation of *PENK* in tissues and blood samples from patients with different cancers, including colorectal, liver, prostate, and oral cancers[17–20]. Circulating DNA may also be excreted transrenally into urine, raising concerns regarding potential false-positive results in urothelial carcinoma diagnosis[21].

To address these concerns, urinary *PENK* methylation levels must be investigated across a broader spectrum of cancers, including UTUC. Accordingly, this study sought to evaluate the urinary *PENK* methylation test across various cancer types to determine its diagnostic accuracy and the potential for false-positive results in urothelial carcinoma diagnosis, with a particular focus on its utility in UTUC detection.

Methods
Study design
This prospective, single-institutional study received Institutional Review Board (IRB) approval from Asan Medical Center (Seoul, South Korea) (IRB number: 2022–0704). All participants provided written informed consent, and the study was carried out in accordance with local ethics guidelines. The study comprised patients diagnosed with genitourinary cancers (bladder cancer, UTUC, prostate cancer, kidney cancer) and non-genitourinary cancers (colon cancer, gastric cancer, liver cancer, esophageal cancer, pancreatic cancer, bile duct cancer, breast cancer, cervical cancer, lung cancer). Eligible participants were 19 to 79 years old, diagnosed with one of the specified cancers between August 2022 and April 2023, and did not receive any treatment (surgery, chemotherapy, hormone therapy, or radiation therapy) after their cancer diagnosis.

Diagnoses were made radiologically or pathologically according to the established diagnostic criteria for each organ-specific cancer. For patients diagnosed at other hospitals, radiological and histopathological examinations were reviewed by our institution's radiologists and pathologists to confirm the diagnoses. Individuals who were diagnosed with two types of cancer simultaneously, had received any treatment before providing specimens, had inadequate specimens, or had any condition that should preclude their participation in the study, as determined by the investigator, were excluded from the study.

All patients underwent diagnosis and treatment according to the protocols for their diagnosed cancers. Consequently, only patients with bladder cancer, UTUC, and cervical cancer underwent cystoscopy. The presence of bladder cancer in other patients was determined using abdomen and pelvic computed tomography (APCT). The study was conducted using a single-blind design, with blinding initiated at the time of urine collection and removed after completion of the methylation analysis. The results of the urinary *PENK* methylation test were then independently analyzed using the cancer types and characteristics of the patients. The flow of this study is depicted in Fig. 1.

Laboratory procedures
Before cystoscopy or definitive treatment, a tube containing 20 mL of freshly voided urine was collected using the Urine Collection Kit I (Genomictree, Inc., Daejeon, South Korea). Samples were centrifuged at 3,000 × g for 10 min to sediment the urine pellets, which were then stored at −20 °C until DNA extraction. Genomic DNA was isolated from the sediments of 10-mL urine using the GT NUCLEIC ACID PREP Kit (Genomictree, Inc.). The extracted DNA was then subjected to sodium bisulfite treatment using the EZ DNA Methylation-Gold kit (Zymo Research, CA, USA) and purified for immediate use in the methylation analysis.

The methylation status of the *PENK* gene was evaluated using the Earlytect BCD assay, a highly sensitive one-step *PENK* methylation (me*PENK*)-linear target enrichment (LTE)-quantitative methylation-specific polymerase chain reaction (qMSP) test[14]. The test used bisulfite-converted DNA obtained from the sediment of 10-mL urine samples. For the control target, a primer set specific to a CpG-free region of the *COL2A1* gene was used. Each reaction mixture contained primers and probes specific for *PENK* methylation, *COL2A1* gene primers and probes, and a universal tag sequence primer. Real-time polymerase chain reaction (PCR) was conducted using a 7500 FAST Dx Real-Time PCR system (Thermo Fisher Scientific, MA, USA).

The *PENK* methylation test was performed once for each sample, and the relative level of the methylated *PENK* gene was calculated as 35-ΔC_T [C_T of amplified *PENK* target – C_T of *COL2A1* (human reference gene)][13,14]. Samples were considered positive if the value of 35-ΔC_T was higher than 32.0 and negative if less than 32.0. A C_T value less than 33.0, which corresponds to the limit of detection for the *COL2A1* gene, was required to ensure assay validity.

Statistical analysis
Patient demographics are presented as continuous variables using mean, median, and range. Categorical variables are presented using frequency and percentile. Sensitivity and specificity were calculated using a dichotomous

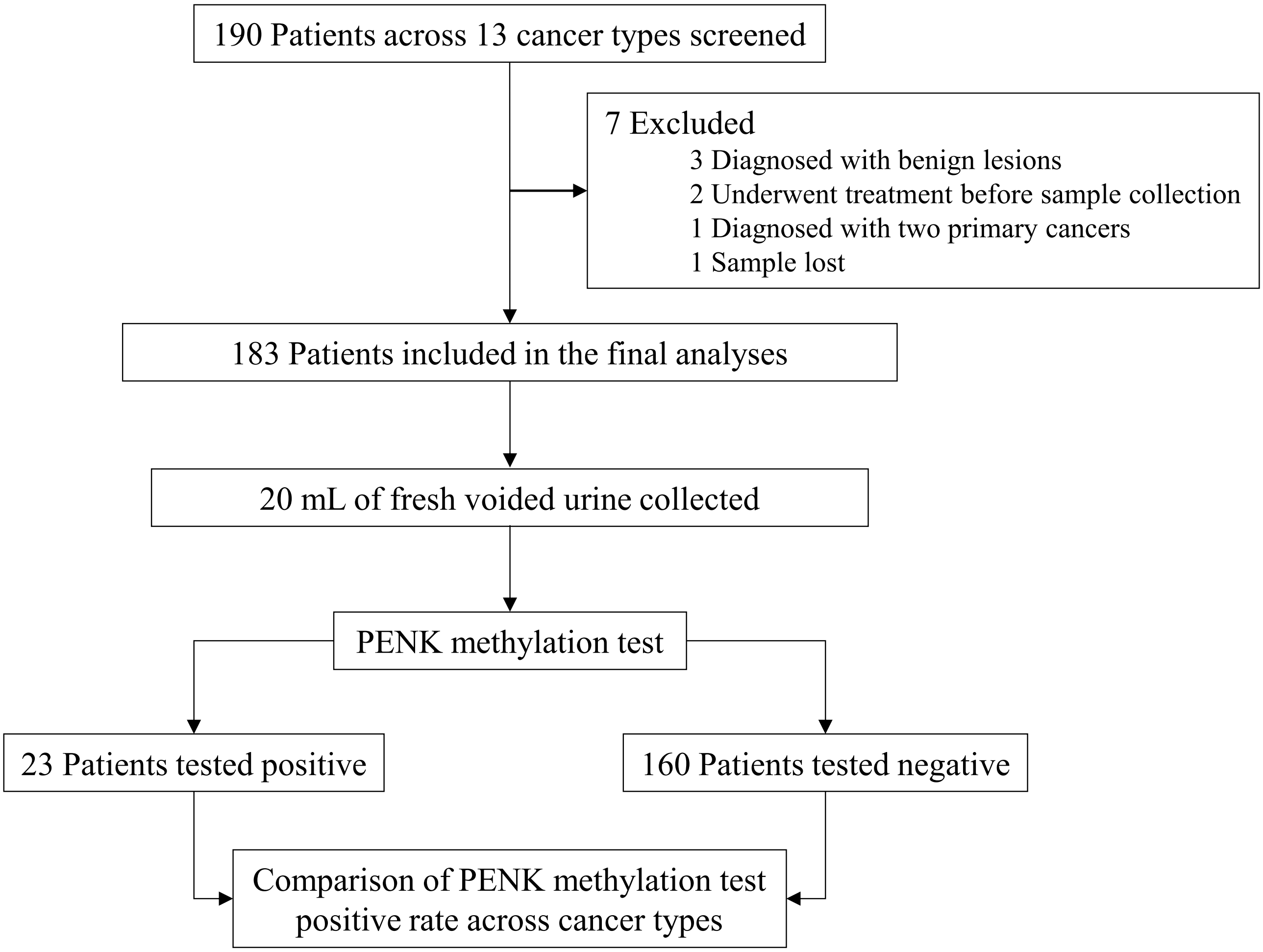

Fig. 1. Flowchart of patient inclusion and *PENK* methylation test results.

approach; patients with urothelial carcinoma were defined as the disease group, and patients with other cancers were defined as the control group. Sensitivity was specifically analyzed for bladder cancer and UTUC separately. In addition to dichotomous classification of test results, we analyzed the distribution of 35–ΔCT values as a semi-quantitative surrogate for *PENK* methylation level. We summarized the median, interquartile range, and range of values for each cancer type. All statistical analyses were performed using R Statistical Software (version 4.3.2, R Core Team, Vienna, Austria).

Results

A total of 190 patients with various stages of 13 different cancer types were prospectively screened. Of these, 7 were excluded, and 183 were included in the final analysis. Detailed reasons for the exclusions are outlined in Fig. 1. The demographic characteristics of the study population are presented in Table 1. The mean age of the participants was 62.3 years, and 72.1% of the participants were males. In terms of smoking status, 39.3% of the participants were nonsmokers, 38.8% were former smokers, and 21.9% were persistent smokers. In addition, 32.8% of the participants had a first-degree relative with cancer. The distribution of cancer types among the study participants is shown in Table 2.

The positivity rates of the urinary *PENK* methylation test across different cancer types are shown in Fig. 2. In the bladder cancer group, 87.5% (7/8) of participants tested positive, while in the UTUC group, 100% (9/9) had positive results. The positivity rate in the control group was 4.2% (7/166). In fact, a positive urinary *PENK* methylation test was obtained for 2 patients with cervical cancer (25.0%), 2 with colorectal cancer (10.5%), and 1 patient each with esophageal (4.8%), liver (5.0%), and kidney (5.3%) cancer.

In a subgroup analysis, we examined the distribution of 35–ΔCT values across cancer types. Patients with urothelial carcinoma exhibited the highest methylation levels, with median values of 36.4 for UTUC and 34.8 for bladder cancer. Cervical cancer, which showed a 25.0% positivity rate, had a median value of 30.9. Most other cancers had median values well below the positivity threshold of 32.0. A detailed summary of the quantitative distribution of 35–ΔCT values by cancer type is provided in Supplementary Table S1. All patients with UTUC and cervical cancer underwent cystoscopy, which confirmed the absence of tumor suspicious lesions within the bladder. Among patients in the non-urothelial carcinoma group that had a positive urinary *PENK* methylation

Characteristic	
Age	
Mean-yr	62.3 ± 9.8
Median (Range)-yr	65 (31—78)
Age group no. (%)	
30–39 yr	5 (2.7)
40–49 yr	16 (8.7)
50–59 yr	40 (21.9)
60–69 yr	69 (37.7)
70–79 yr	53 (29.0)
Sex no. (%)	
Male	132 (72.1)
Female	51 (27.9)
Smoking status no. (%)	
Nonsmokers	72 (39.3)
Former smokers	71 (38.8)
Persistent smokers	40 (21.9)
Cancer in first-degree relatives no. (%)	60 (32.8)

Table 1. Demographic characteristics of the participants.

Cancer type	Number of patients
Bladder cancer	8
Renal pelvis & Ureter cancer	9
Prostate cancer	20
Kidney cancer	19
Esophageal cancer	21
Gastric cancer	19
Colon cancer	19
Liver cancer	20
Pancreatic cancer	10
Bile duct cancer	10
Lung cancer	10
Breast cancer	10
Cervical cancer	8

Table 2. Distribution of cancer types among study participants.

test result, one with colorectal cancer was confirmed to have a renal stone and underwent ureteroscopic stone removal; no evidence of a tumor in the bladder was found at the time of operation. The remaining four patients did not have hematuria based on urinalysis, and no suspicious abnormalities were observed within the urinary tract using APCT.

The urinary *PENK* methylation test had a sensitivity of 94.1% (95% CI: 71.3–99.9%) for detecting urothelial carcinoma; its sensitivity rates were 87.5% for bladder cancer and 100% for UTUC. The test also had a specificity of 95.8% (95% CI: 91.5–98.4%). The bladder cancer group comprised patients with disease ranging from stage Ta low-grade tumors to stage T2. In the UTUC group, the disease stages ranged from T1 to T4, and all tumors were confirmed as pathologic high-grade. Among the patients with bladder cancer and UTUC, none exhibited nodal or distant metastasis.

Urine cytology tests were conducted for 7 patients in each group (bladder cancer and UTUC). In the bladder cancer group, 4 individuals (57.1%) were positive, whereas in the UTUC group, 1 individual (14.3%) was positive. The only patient with a negative urinary *PENK* methylation test result in the bladder cancer group was diagnosed with a stage T1 high-grade tumor and had a positive urine cytology test result.

Discussion

This study aimed to evaluate the diagnostic accuracy of the urinary *PENK* methylation test across different cancer types, focusing on its utility in detecting urothelial carcinoma, including bladder cancer and UTUC. Based on our findings, the urinary *PENK* methylation test has high sensitivity and specificity for detecting urothelial carcinoma, reinforcing its potential as a noninvasive diagnostic tool.

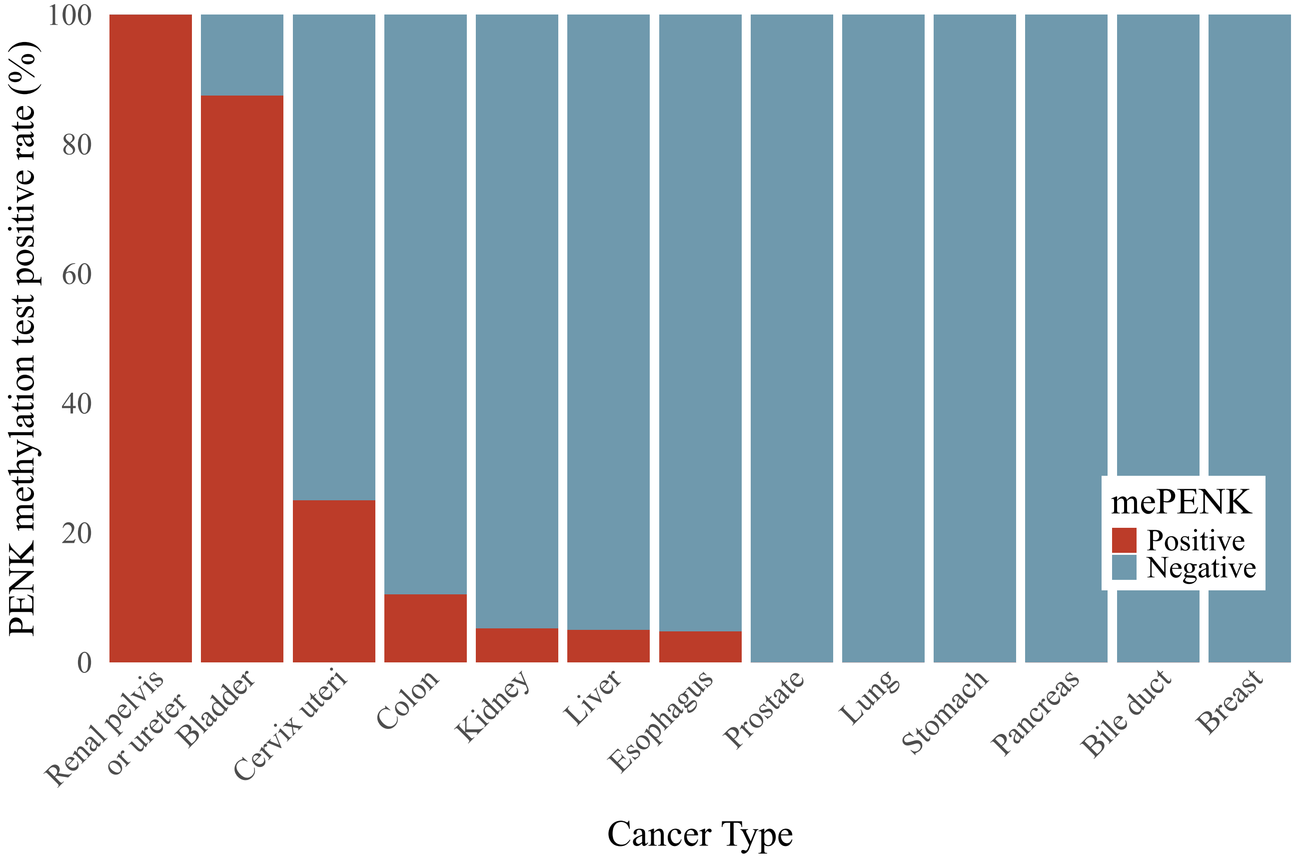

Fig. 2. Rates of positive urinary *PENK* methylation test results across various cancer types.

Several molecular markers for the diagnosis of bladder cancer, such as nuclear matrix protein 22 (NMP22) and UroVysion, are approved by the U.S. FDA and are commercially available. However, these markers have not demonstrated sufficient performance to replace cystoscopy. The NMP22 test has limited diagnostic performance due to its low sensitivity and susceptibility to interference from benign conditions. A systematic review and meta-analysis reported a pooled sensitivity of 56% and specificity of 88%[22]. Although unaffected by benign conditions, UroVysion still has limited diagnostic performance, with a sensitivity of 72% and specificity of 83%[23].

In recent years, novel urinary biomarker tests have been introduced for bladder cancer detection[24]. These advancements hold promise for the diagnosis of bladder cancer and may significantly reduce the reliance on cystoscopy for diagnosis. However, comprehensive data on their diagnostic accuracy for initial bladder cancer detection remains limited. Moreover, many of these tests use multiple DNA or mRNA biomarkers, presenting challenges in terms of reproducibility and high costs[8,25,26].

Previous studies have highlighted the potential of the urinary *PENK* methylation test for bladder cancer detection[12,13,16,27]. A recent large-scale multicenter study involving 1,099 patients with hematuria demonstrated that a urinary DNA methylation test achieved a sensitivity of 89.2% and a specificity of 87.8% for detecting high-grade or invasive bladder cancer, outperforming conventional urinary markers such as NMP22 and urine cytology[16]. In our study, despite including patients with a diverse range of cancers in the control group, the urinary *PENK* methylation test was found to have a sensitivity of 87.5% and a specificity of 95.8% for bladder cancer, consistent with the results of previous studies.

Although bladder cancer and UTUC arise from the same urothelial lining, studies have shown that they differ in genomic profile, clinical behavior, and oncologic outcomes[28,29]. Most studies that use urinary biomarkers for UTUC comprise small and heterogeneous study cohorts, with sensitivity and specificity issues or high inter-observer variability due to subjective evaluation and the need for extensive experience[30]. Our findings are particularly noteworthy in this context.

In its first application to patients with UTUC, the urinary *PENK* methylation test had a sensitivity of 94.1% for overall urothelial carcinoma; 87.5% for bladder cancer, and 100% for UTUC. Moreover, the specificity of the test was 95.8%. It is essential to consider the possibility of UTUC in patients who test positive with the urinary *PENK* methylation test yet have no tumor-like lesions found in exams such as cystoscopy. Given the expanding use of kidney-sparing treatments for patients with UTUC[31], the *PENK* methylation test could play a vital role in the surveillance of these patients, aiding in the early detection of recurrence and progression.

The present study evaluated *PENK* methylation status only in urine, and corresponding tumor tissue methylation profiles were not assessed. However, previous studies have reported *PENK* hypermethylation in various non-urothelial cancers. Specifically, *PENK* gene hypermethylation has been identified in both tumor tissue and blood-derived DNA of colorectal cancer patients[17], in tumor tissue of liver cancer[18], prostate cancer[19], and oral cancer[20]. These observations indicate that *PENK* hypermethylation may occur in various malignancies beyond urothelial carcinoma. This supports the rationale for including a broad spectrum of cancer types in our study.

Two cervical cancer patients in our cohort tested positive for urinary *PENK* methylation, with 35–ΔCT values of 32.4 and 39.0. The two positive patients were classified as pT1, stage I, without evidence of local invasion. *PENK* hypermethylation has not been reported in cervical cancer tissue, and the source of *PENK* methylation detected in urine remains unclear. The anatomical proximity of the cervix to the urinary tract may contribute to local DNA release into urine, and transrenal excretion of circulating methylated DNA may also be involved. Further investigation is needed to clarify the origin and clinical relevance of these findings.

Despite these promising results, several limitations should be acknowledged. First, the patient distribution was not population-based, and the number of patients with urothelial carcinoma was relatively small (n = 17), which limits the generalizability of the reported diagnostic performance. Second, the specificity of 95.8% was calculated using patients with other cancer types as the control group; while this approach demonstrates the test's discriminatory capacity, it may not reflect specificity in an actual screening population. However, as methylation at specific gene sites occurs in various cancers, this study is pivotal for the clinical application of the urinary *PENK* methylation test. Third, although false-positive results were infrequent, the positivity rate was notably higher in cervical cancer patients, which warrants cautious interpretation given the small sample size for this group. Additionally, we did not systematically evaluate gross hematuria in the non-urothelial cancer group; tissue-level *PENK* methylation or post-treatment follow-up data were not collected. These factors limit our ability to fully assess the test's specificity, biological relevance across tumor types, and longitudinal clinical utility. Finally, no additional urothelial evaluation, such as cystoscopy, was performed in patients with non-urothelial cancers who tested positive, as there were no accompanying clinical findings suggestive of urothelial carcinoma. This decision reflects real-world clinical practice but may also have limited the opportunity to identify potential occult UC cases in this subgroup.

Nevertheless, this study had several strengths. The prospective design and inclusion of a diverse range of cancer types enhanced the generalizability of the findings. Furthermore, the single-blind design minimized bias in evaluating the urinary *PENK* methylation test results. Given the increasing evidence supporting urinary DNA methylation biomarkers, including recent large-scale validation studies[16], our findings contribute to the growing body of literature advocating for their clinical implementation.

Conclusions

This study confirmed the high sensitivity and specificity of the urinary *PENK* methylation test for urothelial carcinoma among various cancer types and suggests that the test could be a useful non-invasive biomarker for diagnosing urothelial carcinoma, including bladder cancer and upper tract urothelial carcinoma. This study lays the groundwork for the future clinical application of the urinary *PENK* methylation test in the non-invasive diagnosis of urothelial carcinoma. However, larger-scale prospective studies are required to confirm these findings and optimize the clinical application of the urinary *PENK* methylation test.

Data availability

The data used to support the findings of this study are available from the corresponding author upon reasonable request.

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Author contributions

Research conception and design: Bumjin Lim and In Gab Jeong Data acquisition: Sangmin Lee, In Gab Jeong. Statistical analysis: Sangmin Lee, Bumjin Lim, and Jungyo Suh. Data analysis and interpretation: Sangmin Lee and Jungyo Suh. Drafting of the manuscript: Sangmin Lee and In Gab Jeong. Critical manuscript revision for scientific and factual content: Jungyo Suh, Bumsik Hong, and In Gab Jeong. Administrative, technical, or material support: Bumjin Lim, Jungyo Suh, and In Gab Jeong. Supervision: In Gab Jeong. Approval of the final manuscript: all authors.

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Declarations

Competing interests

The authors declare that they have no competing interests.

Ethics approval

This study was conducted in accordance with the principles outlined in the Declaration of Helsinki and approved by the Institutional Review Board of Asan Medical Center, Seoul, Korea (IRB number: 2022–0704). Written informed consent was obtained from all participants prior to enrollment.

Additional information

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